

Course Code:
CSA06

CO2-Manipulation (BL1, BL2, BL3)

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING) Department of Computer Science and
Engineering ASSESSMENT TOOL 1-ANALYTICAL PROBLEM
SOLVING**

CO Assessed:
Design and Analysis of Algorithms Assessment Tool 1 - Analytical Problem Solving
Course Title:

Assessment :

Weightage:
40% of CIA
Total Marks:
100

CO2 Statement:
Apply Brute Force **technique** to solve **sorting**, searching, Closest-Pair and
Convex-Hull problems (BL1, BL2, BL3)

Student Name :

Section / Batch:
Reg. No.:

Date:

Instructions to Students

- Answer ALL questions. Total Marks: 100.

Analyze each problem carefully before applying the appropriate Brute Force

technique

- **Clearly show all intermediate steps, comparisons, iterations, and logical reasoning** used in solving the problems.
- Apply **suitable algorithms** for sorting, **searching, Closest-Pair, and Convex-Hull** problems wherever required.

Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.

- **Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.**
- **Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning** throughout the answers.

Question Type

Application-**Based** Questions

Questions

Marks

Apply Brute Force techniques for sorting, **searching, Closest-Pair and Convex-Hull** problem analysis

Q1 Q2, Q3

40

60 (2X30)

GRAND TOTAL:

100 Marks

SECTION A- APPLICATION BASED QUESTIONS

Code of Conduct

(192311181)

4 . Party Rakshitha (Name/Reg No) **certify** that this submission is **my original work and that I have**

adhered to the **guidelines** specified for this assessment. I understand that any violation of academic integrity **rules** will result in disciplinary action.

Signature of the Student:

RUBRICS FOR CALCULATION

Criteria

Wt.

Level 4-Exemplary

Level 3- Proficient

Level 2- Developing

Understanding of

20%

Problem Context

Clearly understands the problem and accurately identifies all requirements

Good

understanding with minor gaps

Accurately applies appropriate concepts to solve complex problems

Applies relevant concepts correctly with **minor** errors

Application of

30%

Concepts

Problem-Solving

20%

Approach

Logical, well-structured, and efficient approach

Mostly logical approach with minor gaps

Accuracy of Solution

Completely correct

20%

solution with proper justification

Clear, well-organized,

Clarity

10%

and properly explained solution

Mostly correct with minor errors

Understandable with minor clarity issues

Basic understanding: some requirements unclear

Applies basic

concepts with limited accuracy

Basic approach with

some

Inconsistencies

Partially correct solution

Limited clarity and explanation

Level 1- Beginning

Poor

understanding or misinterpretation of the problem

Fails to apply appropriate concepts

Illogical or
unclear approach

Incorrect or incomplete
solution

Poorly presented and difficult to understand

Marks Evaluation:

Question No.

Understanding

Application of

of Problem

Concepts

Context (20%)

(30%)

Proble... Accuracy

Solving

of **Approach Solution**

(20%) (20%)

Clarity **(10%)**

Total Marks

Q1

4

4

4

4

4

140 Marks)

40

Q2

4

4

4

4

4

30

(30 Marks)

Q3

4

3

3

2

2

22

(30 Marks)

Overall **Total** (100)

92

Faculty **In-charge**

Course Coordinator

Head **of** Department

Signature:

Date:

Signature:

Date:

Signature:

Date:

Brute Force `convex` Hull Algorithm

Given set of points is:

$S = \sqrt{(0,1,3), (2,2), (1,1), (2,1), (3,0), (0,0)), (3,3)}$

*

Easy Identification
 point
 named as

$A = (0,3)$

$B = (R,R)$

$C = (1/4)$

$D = (RIS)$

$E = (310)$

$F =$
 (010)

$G = (3,3)$

point
 coordinates

P1
(013)

(2,2)
P2

P3
(12)

pH
(2,1)
14

P5

(310)

P6
(010)

P7
(313) .

* A concave Hull is the smallest convex polygon that Encloses all given points.

But force *checks* Every pair of points to Determine whether the line joining them is a boundary edge.

Step 1: Total Line pairs

For $n=7$

$n(n-1)$

2

$7 \times 6 = 21$

R

Step 2:—

Total line pairs = 21

checking Line pairs

Pair

All remaining points

on one side

Hull

edge

(013)-(0,0)

yes

(0,3)-(33)

yes

(313) (310)

yes

(3/01 – (010)

yes

Remaining

NO

X

17 pairs

Step 3: valid convex Hull Edges

Edge:

(010) – (013)

(013) – (313)

(313) – (310)

(310) – (010)

Step 4: convex Hull Boundary The

Boundary is formed by:

$(0,0) \rightarrow (0,3) (3,3) (3,0) (0,0)$

Interior

points: $(1,1)$

• $(2,1)$

• $(1,2)$

* These are inside the hull, so

Step 5: Comparisons

hull, so they are not boundary points.

* Line pairs checked = 21

pair

* Remaining points Checked for
each pair, 5

$21 \times 5 = 105$

Total comparisons: 105.

Step 6- Time Complexity

Case

Complexity

Brute Force Conversion

full

$O(n^3)$

Reason :

- * select Every pair of points $O(n^2)$

points $O(n)$

- * Check all other points $\rightarrow \rightarrow O(n)$

Overall $\rightarrow O(n) \times O(n) = O(n^3)$.

Result :-

Item

Result

Total points

7

Line pairs

21

valid Hull Edges

H

Hull vertices

(0,0), (0,3), (3,3), (3,0)

Interior points

105

(1,1), (1,2), (2,1)

Comparisons

Time complexity

$O(n^3)$

Interpretation:—

The Brute force algorithm

identifies the four outer

Corner points as the Convex hull. All

other points lie inside the

boundary. Although simple to

implement, it performs

many comparisons

making it suitable only for small

datasets.